

Yingjie Zhang

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Education

- Duke University**, Master of Science in Computer Science Aug. 2025 – May. 2027
- GPA: 3.67; Coursework: Introduction to Deep Learning (A), Data Science (A-)
- Shanghai Jiao Tong University**, Bachelor of Engineering in Computer Science and Technology Sep. 2021 – Jun. 2025
- GPA: 87.41/100; Coursework: Natural Language Processing (97), Computer Graphics (91)

Internship

- Software Engineer**, Alibaba Group (Amap) – Xiamen, China May. - Aug. 2025
- Built a **multi-agent** map theme generation system on the **Autogen** framework with **Swarm**-based orchestration, leveraging **multimodal LLMs** (e.g., Gemini, GPT-4); ranked **3rd** in the Agent Show competition.
 - Designed and implemented a high-level intent recognition agent to accurately capture user intent, incorporating advanced **prompt engineering** techniques applied across the agent ecosystem.
 - Developed a frontend color-palette workflow embedded in the Theme Color Generation agent via **Autogen Studio (TypeScript/React)**.
- Machine Learning Engineer**, Meituan – Shanghai, China Jun. - Sept. 2024
- Developed and implemented a multilingual text matching system with **SQL**-based data extraction, focusing on detecting merchant self-evaluation patterns across review platforms with a user base of **700M+**.
 - Compressed the 120B **Longcat-Prime** model into 7B **Qwen2-7B-Instruct** by building an efficient **knowledge distillation** pipeline, leveraging **supervised fine-tuning** and custom labeling for optimized transfer.
 - Improved matching precision by **6.14** percentage points (from 91.2% to 97.34%) over the existing regex-based solution and maintained inference efficiency to support **12k+ daily model calls**.
 - Coordinated and guided collaboration between the NLP and Friday LLM platform teams.

Projects & Research Experiences

- Benchmarking Large-Scale EEG Models for Downstream Recognition Tasks — ICASSP 2026** Mar. – Aug. 2025
- Designed and conducted a systematic **benchmark** of **large-scale EEG models** with different **self-supervised pretraining strategies** across multiple downstream recognition settings.
 - Compared five representative **Transformer**-based models (EEGPT, BENDR, BIOT, LaBraM, Gram), uncovering model-task alignment effects and performance degradation under mismatched continued pretraining.
- Meditation EEG Signal Analysis with Self-Supervised Learning — BIBM 2024** Mar. – Sep. 2024
- Proposed and implemented a novel **Multi-view Spectral-Spatial-Temporal Masked Autoencoder (MV-SSTMA)** for EEG representation learning.
 - Achieved consistent performance improvements over seven baselines, with **+6.49% F1** in subject-dependent and **+5.86% F1** in cross-subject evaluations.
- Knowledge Distillation for Speech Recognition Model Whisper (🔗GitHub)** Mar. – Jun. 2024
- Implemented knowledge distillation techniques to optimize OpenAI's Whisper model for Chinese **speech recognition** tasks, leveraging pseudo-labeling for training and **AdaLoRA** for fine-tuning.
 - Achieved substantial improvements through model compression, resulting in a 39% parameter reduction and 2.89x inference speed improvement, while enhancing accuracy with a 3.16% CER reduction.

Skills

Languages: Python, C/C++, SQL

ML & Systems: PyTorch, TensorFlow, HuggingFace, LangChain, FastAPI, Docker, Linux, Git